

ADITI SARKER

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🐙 <https://github.com/krishnaaditi> [Google Scholar Aditi](https://scholar.google.com/citations?user=Aditi)

Research Interests

My research focuses on Multimodal Large Language Models (MLLMs), particularly in hallucination mitigation, modality alignment, and scalable learning. Currently, I am developing a Hierarchical Fine-Tuning approach to enhance modality fusion and reduce errors. My future work includes Direct Preference Optimization (DPO) for improved optimization and adversarial attack analysis to strengthen MLLM security and robustness.

Education

Wayne State University

Aug. 2023 – Present(3.95 out of 4.00)

PhD in Computer Science

Detroit, MI, USA

Ph.D. Candidate (Expected 2028)

- Qualifying Exam (Survey Report) Title: The Role of Contrastive Learning in Multimodal Large Language Models (MLLMs).[Report]

Jahangirnagar University

Apr. 2017 – Nov. 2018(4.00 out of 4.00)

MSc. in Computer Science and Engineering

Dhaka, Bangladesh

- Thesis Title: Simulation of hazy image and validation of haze removal technique.

Jahangirnagar University

Dec. 2011 – Dec.2016(3.78 out of 4.00)

BSc.in Computer Science and Engineering

Dhaka, Bangladesh

Work Experiences

Graduate Research Assistant

May 2024 – August 2024

Wayne State University

Detroit, Michigan

- Developed an algorithm on Hybrid Federated Learning and experiment.
- Developed algorithm on Federated Compositional Optimization and experiment.

Graduate Teaching Assistant

August 2023 – May 2024

Wayne State University

Detroit, Michigan

- Course: Problem Solving and Programming.
- Took lab class, tough programming in c++ language, grading, consultation, exam taking.

Assistant Professor

December 2022 – August 2023

Comilla University

Cumilla Bangladesh

- Course: Data Mining, Artificial Intelligences, Algorithm Analysis and Design, Theory of Computation , Software Engineering.
- Took classes, Took lab classes, grading, consultation, exam taking.

Lecturer

December 2019 – December 2022

Comilla University

Cumilla, Bangladesh

- Course: Graph Theory, Object Oriented Programming, Deep Learning.
- Took classes, Took lab classes, grading, consultation, exam taking.

Fellowships and Awards

Thomas C. Rumble Fellowship

August 2024 – Present

Wayne State University

Ongoing Research

Hallucination-Free Multimodal LLMs: A Unified Training Approach | *Python*

January 2025-Present

- Developed a Hierarchical Fine-Tuning Approach

Publications

Submitted to CVPR 2025 (not published yet)

- **Sarker, A.**, Sultan, R.I., Zhu, H., Zhu, D., Khanduri, P. (2025). Hallucination Mitigation for Large Vision Language Models via Implicit Feature Stabilization and Hierarchical Alignment Optimization. [arxiv link]

Submitted to ICML 2025 (not published yet)

- Khanduri, P., Li, C., Sultan, R.I., **Sarker, A.**, Qiang, Y., Kliwer, J. and Zhu, D. (2025). FEDERATED COMPOSITIONAL OPTIMIZATION: THE IMPACT OF TWO-SIDED LEARNING RATES ON COMMUNICATION EFFICIENCY. [arxiv link]

Journal Articles

- **Sarker, A.**, Chakraborty, P., Sha, S. S., Khatun, M., Hasan, M. R., & Banerjee, K. (2020). Improved technique for analyzing data and detecting terrorist attack using machine learning approach based on twitter data. *Journal of Computer and Communications*, 8 (7), 50–62. doi:10.4236/jcc.2020.87
- **Sarker, A.**, Akter, M., & Uddin, M. S. (2019). Simulation of hazy image and validation of haze removal technique. *Journal of Computer and Communications*, 7 (2), 62–72. doi:10.4236/jcc.2019.72005

Conference Proceedings

- Chandra Das, S., **Sarker, A.**, Saha, S., & Chakraborty, P. (2022). Covid-19 in bangladesh: An exploratory data analysis and prediction of neurological syndrome using machine learning algorithms based on comorbidity. In V. Skala, T. P. Singh, T. Choudhury, R. Tomar, M. Abul Bashar (Eds.), *Machine intelligence and data science applications* (pp. 595–608). doi:10.1007/978-981-19-2347-0
- Uddin, M. S., Gautam, B., **Sarker, A.**, Akter, M., & Haque, M. R. (2017). Image-based automated haze removal using dark channel prior. In 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC) (pp. 412–415). doi:10.1109/R10-HTC.2017.828
- Feroz, M. A., Sultana, M., Hasan, M. R., **Sarker, A.**, Chakraborty, P., & Choudhury, T. (2022). Object detection and classification from a real-time video using ssd and yolo models. In A. K. Das, J. Nayak, B. Naik, S. Dutta, D. Pelusi (Eds.), *Computational intelligence in pattern recognition* (pp. 37–47). Singapore: Springer Singapore.
- Chakraborty, P., Roy, S., Sumaiya, S. N., & **Sarker, A.** (2023). Handwritten character recognition from image using cnn. In *Micro-electronics and telecommunication engineering: proceedings of 6th ICMETE 2022* (pp. 37–47). Singapore: Springer Nature Singapore.
- Chakraborty, P., Yousuf, M. A., Islam, S., Khatun, M., **Sarker, A.**, & Rahman, S. (2022). Analysis of Texture Feature Extraction Technique in Image Processing. In *Distributed Sensing and Intelligent Systems: Proceedings of ICDSIS 2020* (pp. 651–665). Cham: Springer International Publishing.

Books and Chapters

- Uddin, M. S., Shorif, S. B., & **Sarker, A.** (2021). Role-framework of artificial intelligence in combating the covid-19 pandemic (M. A. R. Ahad A. Inoue, Eds.). doi:10.1007/978-3-030-75490-7

Extended Abstract

- **Sarkar, A.**, Rahman, M., & Uddin, M. S. (2020). Automated detection of plan diseases using r- cnn algorithms. *IEEE CSBC Winter Symposium*.
- Sheikh, R., **Sarkar, A.**, Uddin, M. S. (2020). Simulation of haze on real world images with generative adversarial network. *IEEE CSBC Winter Symposium*.

Projects(PhD)

What Can Transformers Learn In-Context? | *Python, C++*

January-April 2024

- The project demonstrate that Transformers can be trained to master more intricate function classes, such as sparse linear functions, decision trees, and two-layer neural networks, achieving performance levels that match or surpass those of task-specific learning algorithms.

CARLA Self-Driving Motion Planning in Autonomous Driving | *Python, C++*

January-April 2024

- We employ a functional motion planning stack. This is accomplished using behavioral planning logic, path selection, velocity profile development, and static collision checking. This project develops a functional motion planning stack for self-driving cars that allows them to avoid both static and dynamic barriers while tracking the center line of a lane and handling stop signs.

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL

Developer Tools: VS Code, Eclipse, Google Cloud Platform, Android Studio

Technologies/Frameworks: Linux, Jenkins, GitHub, JUnit, WordPress

Certified Courses

Advance Machine Learning and Data Science

2020

AILABS, HR Venture

Duration: 6 month

Leadership / Extracurricular

CSE Society

August 2020 – August 2023

Member

Comilla University

Exam Committee

August 2020 – August 2023

Member

Comilla University

IWCI Workshop

2016

Organiser

Jahangirnagar University

References

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Department of Computer Science

Wayne State University, MI

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